

E-commerce Sales Analysis

Step-by-Step Data Analytics Project Plan

Project Goal: Understand where customers buy more, what they buy, how much they spend, and what factors may influence their purchasing behavior.

Phase 1 — Understand the Dataset

Step 1 — Understand every column

- Identify what each column represents.
- Classify each column as numerical, categorical, date, or ID.
- Write down what business question each column can help answer.

Step 2 — Check data quality

- Check for missing values, duplicate orders, incorrect dates, impossible values, outliers, and incorrect data types.
- Goal: make sure the dataset is trustworthy before analysis.

Phase 2 — Get a Basic Business Overview

Step 3 — Calculate the main KPIs

- Total revenue
- Total orders
- Total customers
- Total quantity sold
- Average order value
- Average customer rating
- Average delivery time

Step 4 — Answer the first business question

- How is the overall e-commerce business performing?
- Explain what the numbers mean instead of only reporting them.

Phase 3 — Analyze Sales Over Time

Step 5 — Study the order dates

- Analyze orders, revenue, and quantity over time.
- Start with monthly analysis.

Step 6 — Find the best and worst periods

- Identify the highest and lowest revenue months.
- Identify the highest and lowest order-volume months.

Step 7 — Find patterns

- Is the business growing?
- Are there periods when customers buy more?
- Are revenue and order volume moving together?

Phase 4 — Regional Analysis

Step 8 — Compare all regions

- Compare orders, quantity purchased, total revenue, average order value, average rating, average delivery time, and average discount.

Step 9 — Identify the winners

- Find the region with the highest and lowest orders, quantity, revenue, average order value, and rating.

Step 10 — Answer the major regional question

- Which region has the strongest customer purchasing behavior?
- Do not automatically call the region with the most orders the best; compare multiple metrics.

Phase 5 — Product Category Analysis

Step 11 — Compare product categories

- Compare orders, quantity, revenue, average price, average discount, and customer rating.

Step 12 — Find the most popular category

- What are customers buying the most?

Step 13 — Find the most valuable category

- Which category generates the most revenue?
- Check whether the most popular and most valuable categories are different.

Phase 6 — Region × Product Category

Step 14 — Find what each region prefers

- Determine the most-purchased product category within each region.

Step 15 — Compare regional preferences

- Does customer preference change depending on the region?
- This can become one of the strongest findings in the project.

Phase 7 — Customer Spending Analysis

Step 16 — Understand individual customers

- Analyze number of orders per customer, total spending, quantity purchased, and average order value.

Step 17 — Find high-value customers

- Identify top customers by spending, order frequency, and average order value.

Step 18 — Compare customer groups

- Are frequent customers also spending more?
- Are high-spending customers concentrated in certain regions?

Phase 8 — Find “WHY” Customers Buy More

Step 19 — Investigate discounts

- Do customers purchase more when discounts are higher?
- Compare discount with quantity and revenue.

Step 20 — Investigate pricing

- Does product price appear to affect how much customers purchase?

Step 21 — Investigate delivery

- Does delivery speed appear to affect customer satisfaction?

Step 22 — Investigate customer satisfaction

- Do highly rated experiences appear to be associated with more purchasing?

Step 23 — Avoid unsupported conclusions

- Use language such as “the data shows an association” rather than claiming that one factor definitely causes another.
- The dataset does not directly contain a customer-reported reason for purchase.

Phase 9 — Find the Highest Average Value

Step 24 — Compare Average Order Value

- Compare AOV between regions, product categories, payment methods, and time periods.
- Identify which region has the highest average order value.

Step 25 — Put the AOV finding into context

- How much higher is it than the overall average?
- Does the region also have high revenue?
- Does it have fewer or more orders?
- What categories contribute to its high AOV?

Phase 10 — Build Your Business Story

Step 26 — Select the strongest findings

- Choose approximately 10–15 meaningful findings.
- For each finding, document: Finding, Evidence, Possible Explanation, Business Impact, and Recommendation.

Phase 11 — Visualization

Step 27 — Create focused charts

- 1. Monthly Revenue Trend
- 2. Revenue by Region
- 3. Average Order Value by Region
- 4. Revenue by Product Category
- 5. Quantity by Product Category

- 6. Region × Category Heatmap
- 7. Discount vs Quantity
- 8. Delivery Days vs Customer Rating

Step 28 — Make every chart answer a question

- Do not create charts just because the data allows it. Each visualization should support a specific business question.

Phase 12 — Final Conclusion

Step 29 — Answer the major questions

- How much is the business selling?
- When are customers buying more?
- Where are customers buying more?
- Where is average order value highest?
- What are customers buying?
- What does each region prefer?
- Who are the highest-value customers?
- What factors appear to influence purchasing behavior?
- What should the business do based on the findings?

Final Analysis Flow

Understand Dataset → Clean Dataset → Calculate KPIs → Analyze Time → Analyze Regions → Analyze Categories → Region × Category → Analyze Customers → Investigate WHY → Compare AOV → Find Insights → Visualize → Recommend → Final Dashboard/Report

Core question: What happened → Where → Who → What are they buying → Why might it be happening → What should the business do?